
STATE NAME

STATE NAME estimates *(one-sentence definition of what this state measures)*. It is theoretically distinct from *(related state 1)* (which measures X) and from *(related state 2)* (which measures Y). *(One or two sentences on the defining conceptual boundary of this state – what makes it a separate, non-redundant construct.)*

Scope. *State a clear, honest boundary on what this score does and does not claim to measure. Name the specific confound or misinterpretation risk most likely to occur, and state what additional signal or chapter is needed to resolve it.*

1.1 Theoretical Basis

(1–2 paragraphs grounding the formula in published theory. Name the specific framework(s) being used – a named model, a specific physiological mechanism, a specific cognitive theory – and cite the key supporting reference(s) by author and year. State explicitly which parts of the formula below are directly evidenced by this theory, and flag any parts that are engineering adaptations rather than direct citations.)

- **Mechanism or signal 1** — *one-sentence explanation of why this signal is relevant, with citation.*
- **Mechanism or signal 2** — *one-sentence explanation.*

1.2 Input Signals

Signal	Symbol	Description
Signal 1 name	X1	What it is, units, source
Signal 2 name	X2	What it is, units, source
Personal baseline	X _{base}	Rolling personal baseline, recalculation cadence
Activity state	ACT	Sedentary / standing / walking / running
Movement regularity	reg	Kinematic smoothness score $\in [0, 1]$

1.3 Rule Formula

RULE-XX *STATE NAME Score*

Step 1.1 — Sub-computation name:

$$\text{term}_1 = \text{clamp}\left(\frac{X - X_{\text{base}}}{X_{\text{base}}}, 0, 1\right)$$

Example: $X = (\text{value})$, $X_{\text{base}} = (\text{value})$: $\text{term}_1 = \text{clamp}((\text{calc}), 0, 1) = (\text{result})$

Step 1.2 — Sub-computation name:

$$\text{term}_2 = \text{clamp}(\dots)$$

Step 1.3 — Motion gate (if applicable):

$$\text{motion_gate} = \text{base_gate}(\text{ACT}) \times (0.7 + 0.3 \times \text{reg})$$

Step 1.4 — Combine into final score:

$$\boxed{\text{STATE} = \text{clamp}(\dots, 0, 1)} \quad \text{STATE} \in [0, 1]$$

1.4 Parameter Notes

Parameter or term name

What it represents, why this specific design choice was made, and any known limitation or assumption behind it.

Parameter or term name

Same structure.

Fusion method (if applicable)

State whether terms are combined via weighted sum, Noisy-OR, or another method, and justify the choice – e.g. Noisy-OR for genuinely independent evidence paths, weighted average for correlated/non-independent signals measuring the same underlying construct.

1.5 Confidence

Step 2.1 — Confidence factor 1:

$$C_{\text{name1}} = \dots$$

Step 2.2 — Confidence factor 2:

$$C_{\text{name2}} = \dots$$

Step 2.3 — Final confidence:

$$\text{conf} = C_{\text{name1}} \times C_{\text{name2}} \times \dots$$

State any permanent confidence ceiling here, if the formula adapts an unvalidated mechanism, and explain why the ceiling exists.

1.6 Interpretation Scale

Score	Level	Meaning
0.0–0.2	Level 1	Meaning
0.2–0.4	Level 2	Meaning
0.4–0.6	Level 3	Meaning
0.6–0.8	Level 4	Meaning
0.8–1.0	Level 5	Meaning

1.7 Context Applicability

Target context: *State the specific real-world context this formula was designed for and is most reliable in.*

Context	Signal 1	Signal 2	Reliable?
Context A	Yes	Yes	Yes — full
Context B	Partial	Yes	Partial
Context C	No	No	No

Scientific boundary: *State explicitly what this formula cannot claim, even under ideal data conditions – e.g. a confound it cannot disambiguate, a construct it approximates rather than directly measures, or a population/context it has not been tested on.*

1.8 Worked Example: Realistic Scenario Name

Scenario: *2–4 sentences describing a specific, realistic person, time, activity, and context – concrete enough that every input value below is plausible and internally consistent.*

Signal	Value
Signal 1	value
Signal 2	value

Step-by-step calculation:

$$\text{term} = \text{clamp}(\text{calc with actual numbers}, 0, 1) = (\text{result})$$

$$\text{STATE} = (\text{final calc}) = (\text{final result})$$

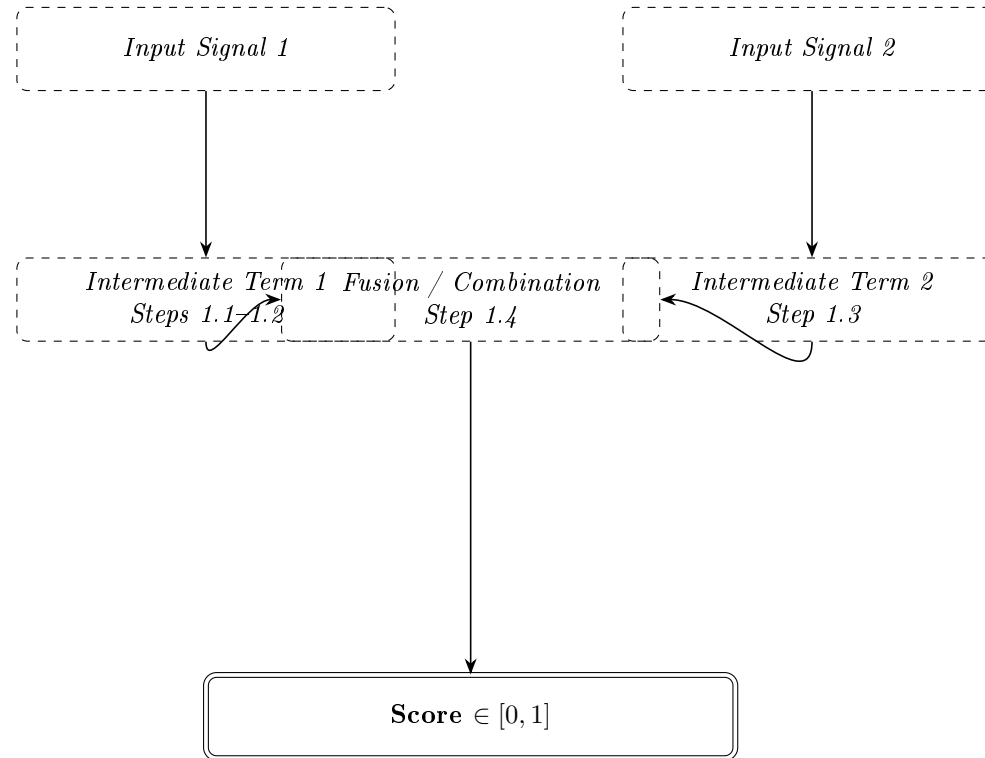
Interpretation: *State what this result means in the context of the scenario, which band it falls into, and what the dominant driver of the score was – name the specific term that contributed most.*

1.9 Implementation Notes

- **Window.** *Window size and step interval used.*
- **Fallback behavior.** *What happens when a required signal is unavailable – does the formula degrade gracefully, and how?*
- **Relationship to other chapters.** *Which other states this should be combined with, and why neither alone is sufficient.*
- **Validation status.** *State plainly whether this formula has been validated on real data, on what dataset, and what remains untested.*

Formula Flow Diagram — Chapter N: STATE NAME

STATE NAME



Replace with full pipeline diagram when formula is finalised.